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FOUNDATIONAL TIME SERIES MODELS FOR EMERGENCY DEPARTMENT OCCUPANCY FORECASTING

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ABSTRACT

Valtti Järvi: Foundational time series models for emergency department occupancy forecasting
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Emergency department overcrowding is a significant threat to patient safety..

Keywords: time series forecasting, foundational models, emergency department occupancy prediction

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TIIVISTELMÄ

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Avainsanat: keyword, keyword, keyword, keyword, keyword

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CONTENTS

1. Introduction	1
2. Background	2
2.1 ED forecasting	2
2.1.1 ED crowding	2
2.1.2 ED forecasting	2
2.1.3 ED time series data	3
2.2 Foundation models	3
2.2.1 Introduction	3
2.2.2 Paradigm shift	4
2.2.3 Model architectures	4
2.3 Related work	4
2.3.1 Current state of the art in ED forecasting	4
2.3.2 Performance of foundational models in other fields	4
3. Methodology	5
3.1 Models	5
3.2 Data	5
3.3 Experimental setup	5
3.4 Evaluation	5
3.5 Finetuning	5
4. Results	6
4.1 Main results	6
4.2 Effect of context length	6
4.3 Effect of covariates	6
4.4 Visualizations	6
5. Discussion	7
6. Conclusion	8
7. References	9

1. INTRODUCTION

Overcrowding in hospital emergency departments (EDs) have a clear negative impact on patients health [1]. For example, a higher mortality rate [2][3] and increased length of stay [4] have been associated with crowding. Accurate predictions of future ED occupancy levels could prevent crowding by allocating resources differently.

Current state of the art (SOTA) machine learning models for ED occupancy prediction are trained by using historical occupancy data and sometimes other external variables, like weather and traffic data. (lähde) Training machine learning models is computationally heavy and time consuming task. Every hospital having unique characteristics, model trained with data from one hospital could possibly not predict well future occupancy levels for other hospitals. Foundational models do not need training to predict from specific data. If they could predict ED occupancy accurately enough, hard part of model training can be skipped. This could save substantial amount of computation and man hours.

In this thesis foundational time series models are evaluated using ED occupancy data from Acuta. Acuta is the Tampere university hospital's (TAYS) emergency department. TAYS is located in Tampere, Finland. The data used is identical to that in a study by Tuominen et al. [5]. In their work, they created machine learning models to predict ED occupancy. This allows for a direct comparison between the performance of foundation models and their results.

The goal of this thesis was to examine how well the foundational time series models perform on emergency department occupancy prediction. In total five different models were chosen and tested. Each model was tested with multiple different input lengths. The models were also tested with two different types of input data. The first type uses only past ED occupancy data. This is called univariate input. The second type uses past ED occupancy data combined with external variables like weather and traffic data. This is called multivariate input.

The thesis is structured as follows. In chapter 2, ED forecasting, foundation models and related research are introduced. Chapter 3 covers the details of how the experiment was conducted. Results are presented in chapter 4. (Chapter 5 analyzes the results and chapter 6 concludes the thesis.)

2. BACKGROUND

2.1 ED forecasting

2.1.1 ED crowding

ED overcrowding does not have a single widely accepted definition. Different studies use different metrics, such as number of arrivals, clinician opinion of crowding or number of full rooms [6]. One verbal definition of overcrowding is "a situation in which demand for service exceeds the ability to provide care within a reasonable time, causing physicians and nurses to be unable to provide quality care." [7].

Most widely used conceptual model is the input-throughput-output model. Input refers to factors related to patient arrivals and demand for ED services. Throughput includes actions inside ED. Output refers to ED patient disposition [8].

ED overcrowding has become a worldwide problem [9]. Its negative consequences are commonly divided into three categories. First, patients health is impacted negatively. Higher mortality rate is clearly associated with crowding several days after hospital visit [2, 3, 10, 11, 12]. Secondly, mistakes by hospital staff are more common [13](tarkista) and stress levels are higher for nurses and physicians [14]. Thirdly, there is a system level disruption. Diverting ambulances to another ED [8] delays patients access to care and disrupts coordination between hospitals. Patients' waiting time for treatment increases substantially [4]. Also, EDs cannot work as a safety net for mass casualty incidents and other patient surges while crowded [9].

2.1.2 ED forecasting

Forecasting overcrowding enables proactive resource allocation to EDs. Accurate predictions can then prevent overcrowding and its harmful consequences. There are different metrics that can be forecasted and have been linked to overcrowding. Forecasting ED patient demand or patient admission are commonly used metrics. They contribute to input part of the input-throughput-output model. Forecasting of throughput, also called length of stay, is also used. It is the time patient spend in ED from arrival to leaving or moving to other department in the hospital. More rarely ED crowding is defined by using some

mathematical formula, such as ratio of total patients to total beds. [15]

ED occupancy is also used as a forecasting metric. All patients from all ED treatment spaces are summed up to this metric. All parts of input-throughput-output model are affected in occupancy, which makes it useful. [5]

In practice, forecasting relies on historical observations to learn recurring patterns like daily, weekly, and seasonal cycles, long-term trend, and short-term autocorrelation. Explanatory variables, such as weather, traffic, and hospital bed availability, can be used to improve accuracy and operational usefulness. Forecasts can be point estimates or full predictive distributions. [5]

2.1.3 ED time series data

A discrete-time time series is a set of discrete values $\{x_t\}$, where t denotes a specific time [16]. In this thesis, the term 'time series' refers always to discrete-time time series. The intervals between t are usually fixed and of equal length.

The word univariate is used to describe data that consists of a single attribute. For example, hourly occupancy data is univariable. Multivariable describes data that consists of multiple independent attributes.(lähde?) It can be denoted as $\{\mathbf{x}_t\}$, where $\mathbf{x}_t$ is a vector [16].

- erityispiirteitä ED aikasarjadatasta, miten ED data käyttäytyy lyhyellä aikavälillä?

2.2 Foundation models

2.2.1 Introduction

A foundation models are an artificial intelligence (AI) systems that performs well at a wide range of diverse tasks. These models are trained on massive amounts of diverse data. Foundation models use deep neural network architectures with parameters ranging from hundreds of millions to hundreds of billions, depending on the application domain. [17] For example, foundation models have been developed for time series forecasting. A single foundation time series model can predict diverse time series data, from ED occupancy to stock prices.

Transfer learning is a concept of transferring knowledge from one learned task to other tasks [**<empty citation>**]. From vast training data, fundamental structures that are true for all tasks in one domain, are learned [**<empty citation>**]. For example, ED occupancy and stock prices could have same kind of fluctuation. Transfer learning and the vast size of foundation models is what makes them work. [17]

In [17], the authors state three advancements that have enabled the size of foundation models. First, improvement in computational hardware. Second, the Transformer architecture that efficiently utilized this hardware for training [18]. Third, access to larger datasets for training.

2.2.2 Paradigm shift

Deep learning is a subset of machine learning that uses deep neural networks to learn representations from data. The networks automatically extract features at different levels of abstraction. [19] In [20] the author explains the key components of creating machine learning models. Deep learning uses this same approach [19]. First step is choosing representation, or model

Foundation models represents a paradigm shift from this traditional approach.

- paradigm from traditional ml model training+validation -> foundational models -In short, differs from traditional ml models in scale, generality, adaptation -consequences: saves computation, time and effort, is generalizable

2.2.3 Model architectures

- short and high level description of architectures and features of each model. naming of each model

[21] sundial [22] chronos [23] tirex [24] timesfm [25] moirai

2.3 Related work

2.3.1 Current state of the art in ED forecasting

- tietoa nykyisestä state of artista ED ennustuksessa: ARIMAX, TFT, lightGBM, seasonal Naive [5] [26] TFT

2.3.2 Performance of foundational models in other fields

- Performance of foundational time series prediction for ED occ in other studies (if any)
- Performance of foundational time series prediction for other data (find studies) [27] survey across all domains

3. METHODOLOGY

3.1 Models

- criteria for choosing models
- settings and hyper-parameters that were used for each model
- very short description of trained ml models that these models are compared against

3.2 Data

- where the data came from
- describe the data used (univariate, multivariate...) and how it was processed
- data allows direct comparison with other ml models trained on the same data.

3.3 Experimental setup

- task is to predict 24h occupancy to the future
- uni vs multivariate inputs
- different context length inputs
- hardware setup and (library versions, date when models were downloaded?)

3.4 Evaluation

- describe the evaluation metrics used (rmse, mae, msis), what they tell and why they are used

3.5 Finetuning

- tämä osio jos finetunausta tehdään

4. RESULTS

4.1 Main results

- show the charts and tables of the main results (rmse, mae, msis)

4.2 Effect of context length

- charts that show error with different context lengths

4.3 Effect of covariates

- chart comparing uni vs multi inputs

4.4 Visualizations

- show single day prediction vs ground truth. (best, worst and typical days)

5. DISCUSSION

- how accuracy changes with input length
- how input covariates changes accuracy
- did fine tuning increase accuracy
- are foundational models potential option for areas like this one, comparison to trained models, compare with other studies: Show chart with all the old results and new results. Compare them and keep telling the story of trained vs foundational
- limitations of the study (only one validation dataset)
- other observations and future work
- maybe connect with results or conclusion chapter?

6. CONCLUSION

- summary of answers to the research questions, conclusion

7. REFERENCES

- [1] Sartini, M., Carbone, A., Demartini, A., Giribone, L., Oliva, M., Spagnolo, A. M., Cremonesi, P., Canale, F. and Cristina, M. L. Overcrowding in Emergency Department: Causes, Consequences, and Solutions—A Narrative Review. *Healthcare* 10.9 (Aug. 25, 2022), p. 1625. ISSN: 2227-9032. DOI: 10.3390/healthcare10091625. (Visited on 09/25/2025).
- [2] Berg, L. M., Ehrenberg, A., Florin, J., Östergren, J., Discacciati, A. and Göransson, K. E. Associations Between Crowding and Ten-Day Mortality Among Patients Allocated Lower Triage Acuity Levels Without Need of Acute Hospital Care on Departure From the Emergency Department. *Annals of Emergency Medicine* 74.3 (Sept. 1, 2019), pp. 345–356. ISSN: 0196-0644. DOI: 10.1016/j.annemergmed.2019.04.012. URL: <https://www.sciencedirect.com/science/article/pii/S0196064419303312> (visited on 09/25/2025).
- [3] Jo, S., Jin, Y. H., Lee, J. B., Jeong, T., Yoon, J. and Park, B. Emergency Department Occupancy Ratio is Associated With Increased Early Mortality. *Journal of Emergency Medicine* 46.2 (Feb. 1, 2014), pp. 241–249. ISSN: 0736-4679. DOI: 10.1016/j.jemermed.2013.05.026.
- [4] McCarthy, M. L., Zeger, S. L., Ding, R., Levin, S. R., Desmond, J. S., Lee, J. and Aronsky, D. Crowding Delays Treatment and Lengthens Emergency Department Length of Stay, Even Among High-Acuity Patients. *Annals of Emergency Medicine* 54.4 (Oct. 1, 2009), 492–503.e4. ISSN: 0196-0644. DOI: 10.1016/j.annemergmed.2009.03.006.
- [5] Tuominen, J., Pulkkinen, E., Peltonen, J., Kanninen, J., Oksala, N., Palomäki, A. and Roine, A. Forecasting emergency department occupancy with advanced machine learning models and multivariable input. *International Journal of Forecasting* 40.4 (Oct. 1, 2024), pp. 1410–1420. ISSN: 0169-2070. DOI: 10.1016/j.ijforecast.2023.12.002. URL: <https://www.sciencedirect.com/science/article/pii/S0169207023001346> (visited on 09/11/2025).
- [6] Hwang, U., McCarthy, M. L., Aronsky, D., Asplin, B., Crane, P. W., Craven, C. K., Epstein, S. K., Fee, C., Handel, D. A., Pines, J. M. et al. Measures of crowding in the emergency department: a systematic review. *Academic Emergency Medicine* 18.5 (2011), pp. 527–538.

- [7] : *Canadian Journal of Emergency Medicine* 3.2 (2001), pp. 82–84. DOI: 10.1017/S1481803500005285.
- [8] Asplin, B. R., Magid, D. J., Rhodes, K. V., Solberg, L. I., Lurie, N. and Camargo Jr, C. A. A conceptual model of emergency department crowding. *Annals of emergency medicine* 42.2 (2003), pp. 173–180.
- [9] Bernstein, S. L., Aronsky, D., Duseja, R., Epstein, S., Handel, D., Hwang, U., McCarthy, M., McConnell, K. J., Pines, J. M., Rathlev, N., Schafermeyer, R., Zwemer, F., Schull, M. and Asplin, B. R. Emergency Department Crowding, Part 1: Definition and Magnitude. *Academic Emergency Medicine* 16.11 (2009), pp. 106–113. DOI: 10.1111/j.1553-2712.2009.00590.x.
- [10] Eidstø, A., Ylä-Mattila, J., Tuominen, J., Huhtala, H., Palomäki, A. and Koivistoinen, T. Emergency department crowding increases 10-day mortality for non-critical patients: a retrospective observational study. *Internal and Emergency Medicine* 19.1 (2024), pp. 175–181.
- [11] Guttman, A., Schull, M. J., Vermeulen, M. J. and Stukel, T. A. Association between waiting times and short term mortality and hospital admission after departure from emergency department: population based cohort study from Ontario, Canada. *BMJ* 342 (2011). ISSN: 0959-8138. DOI: 10.1136/bmj.d2983. eprint: <https://www.bmj.com/content/342/bmj.d2983.full.pdf>. URL: <https://www.bmj.com/content/342/bmj.d2983>.
- [12] Richardson, D. B. Increase in patient mortality at 10 days associated with emergency department overcrowding. *The Medical Journal of Australia* 184.5 (Mar. 6, 2006), pp. 213–216. ISSN: 0025-729X. DOI: 10.5694/j.1326-5377.2006.tb00204.x. URL: <https://www.mja.com.au/journal/2006/184/5/increase-patient-mortality-10-days-associated-emergency-department-overcrowding>.
- [13] Kulstad, E. B., Sikka, R., Sweis, R. T., Kelley, K. M. and Rzechula, K. H. ED overcrowding is associated with an increased frequency of medication errors. *The American Journal of Emergency Medicine* 28.3 (2010), pp. 304–309. ISSN: 0735-6757. DOI: <https://doi.org/10.1016/j.ajem.2008.12.014>. URL: <https://www.sciencedirect.com/science/article/pii/S0735675708008310>.
- [14] Bond, K., Ospina, M., Blitz, S., Afilalo, M., Sam, G., Bullard, M., Innes, G., Holroyd, B., Curry, G., Schull, M. and Rowe, B. Frequency, Determinants and Impact of Overcrowding in Emergency Departments in Canada: A National Survey. *Health-care quarterly (Toronto, Ont.)* 10 (Feb. 2007), pp. 32–40. DOI: 10.12927/hcq.2007.19312.
- [15] Gul, M. and Celik, E. An exhaustive review and analysis on applications of statistical forecasting in hospital emergency departments. *Health Systems* 9.4 (2018). No potential conflict of interest was reported by the authors., pp. 263–284. ISSN:

- 2047-6965. DOI: 10.1080/20476965.2018.1547348. URL: <https://www.tandfonline.com/doi/full/10.1080/20476965.2018.1547348>.
- [16] Peter J. Brockwell, R. A. D. *Introduction to Time Series and Forecasting*. 2016. DOI: 10.1007/978-3-319-29854-2.
- [17] Bommasani, R. et al. *On the Opportunities and Risks of Foundation Models*. 2022. arXiv: 2108.07258 [cs.LG]. URL: <https://arxiv.org/abs/2108.07258>.
- [18] Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A. N., Kaiser, L. and Polosukhin, I. Attention Is All You Need. *Advances in Neural Information Processing Systems*. 2017, pp. 5998–6008. URL: <https://arxiv.org/abs/1706.03762>.
- [19] LeCun, Y., Bengio, Y. and Hinton, G. Deep Learning. *Nature* 521.7553 (2015). Published online: 27 May 2015; Issue date: 28 May 2015, pp. 436–444. DOI: 10.1038/nature14539. URL: <https://www.nature.com/articles/nature14539>.
- [20] Domingos, P. A Few Useful Things to Know About Machine Learning. *Communications of the ACM* 55.10 (2012), pp. 78–87. DOI: 10.1145/2347736.2347755.
- [21] Liu, Y., Qin, G., Shi, Z., Chen, Z., Yang, C., Huang, X., Wang, J. and Long, M. *Sundial: A Family of Highly Capable Time Series Foundation Models*. May 30, 2025. DOI: 10.48550/arXiv.2502.00816. arXiv: 2502.00816. URL: <http://arxiv.org/abs/2502.00816> (visited on 09/11/2025).
- [22] Ansari, A. F., Stella, L., Turkmen, C., Zhang, X., Mercado, P., Shen, H., Shchur, O., Rangapuram, S. S., Arango, S. P., Kapoor, S., Zschiegner, J., Maddix, D. C., Wang, H., Mahoney, M. W., Torkkola, K., Wilson, A. G., Bohlke-Schneider, M. and Wang, Y. *Chronos: Learning the Language of Time Series*. Nov. 4, 2024. DOI: 10.48550/arXiv.2403.07815. arXiv: 2403.07815. URL: <http://arxiv.org/abs/2403.07815> (visited on 09/11/2025).
- [23] Auer, A., Podest, P., Klotz, D., Böck, S., Klambauer, G. and Hochreiter, S. *TiRex: Zero-Shot Forecasting Across Long and Short Horizons with Enhanced In-Context Learning*. May 29, 2025. DOI: 10.48550/arXiv.2505.23719. arXiv: 2505.23719. URL: <http://arxiv.org/abs/2505.23719> (visited on 09/11/2025).
- [24] Das, A., Kong, W., Sen, R. and Zhou, Y. *A decoder-only foundation model for time-series forecasting*. Apr. 17, 2024. DOI: 10.48550/arXiv.2310.10688. arXiv: 2310.10688. (Visited on 09/11/2025).
- [25] Woo, G., Liu, C., Kumar, A., Xiong, C., Savarese, S. and Sahoo, D. *Unified Training of Universal Time Series Forecasting Transformers*. May 22, 2024. DOI: 10.48550/arXiv.2402.02592. arXiv: 2402.02592. URL: <http://arxiv.org/abs/2402.02592> (visited on 09/11/2025).
- [26] Koya, S. R. and Roy, T. Temporal Fusion Transformers for Streamflow Prediction: Value of Combining Attention with Recurrence. *Journal of Hydrology* 637 (2024), p. 131301.

- [27] Jin, M., Wen, Q., Liang, Y., Zhang, C., Xue, S., Wang, X., Zhang, J., Wang, Y., Chen, H., Li, X. et al. Foundation Models for Time Series Analysis: A Tutorial and Survey. *Proceedings of the 30th ACM SIGKDD Conference on Knowledge Discovery and Data Mining*. 2024, pp. 6407–6416.